



Effect of fixed-dose subcutaneous reslizumab on asthma exacerbations in patients with severe uncontrolled asthma and corticosteroid sparing in patients with oral corticosteroid-dependent asthma: results from two phase 3, randomised, double-blind, placebo-controlled trials

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Summary

Background Reslizumab 3 mg/kg administered intravenously is approved for the treatment of severe eosinophilic asthma. We assessed the safety and efficacy of subcutaneous reslizumab 110 mg in two trials in patients with uncontrolled severe asthma and increased blood eosinophils. The aim was to establish whether subcutaneous reslizumab 110 mg can reduce exacerbation rates in these patients (study 1) or reduce maintenance oral corticosteroid dose in patients with corticosteroid-dependent asthma (study 2).

Methods Both studies were randomised, double-blind, placebo-controlled, phase 3 studies. Entry criteria for study 1 were uncontrolled severe asthma, two or more asthma exacerbations in the previous year, a blood eosinophil count of 300 cells per μL or more (including no more than 30% patients with an eosinophil count <400 cells/ μL), and at least a medium dose of inhaled corticosteroids with one or more additional asthma controllers. Patients in study 2 had severe asthma, a blood eosinophil count of 300 cells per μL or more, daily maintenance oral corticosteroid (prednisone 5–40 mg, or equivalent), and high-dose inhaled corticosteroids plus another controller. Patients were randomly assigned (1:1) to subcutaneous reslizumab (110 mg) or placebo once every 4 weeks for 52 weeks in study 1 and 24 weeks in study 2. Patients and investigators were masked to treatment assignment. Primary efficacy outcomes were frequency of exacerbations during 52 weeks in study 1 and categorised percentage reduction in daily oral corticosteroid dose from baseline to weeks 20–24 in study 2. Primary efficacy analyses were by intention to treat, and safety analyses included all patients who received at least one dose of study treatment. These studies are registered with ClinicalTrials.gov, NCT02452190 (study 1) and NCT02501629 (study 2).

Findings Between Aug 12, 2015, and Jan 31, 2018, 468 patients in study 1 were randomly assigned to placebo ($n=232$) or subcutaneous reslizumab ($n=236$), and 177 in study 2 to placebo ($n=89$) or subcutaneous reslizumab ($n=88$). In study 1, we found no significant difference in the exacerbation rate between reslizumab and placebo in the intention-to-treat population (rate ratio 0.79, 95% CI 0.56–1.12; $p=0.19$). Subcutaneous reslizumab reduced exacerbation frequency compared with placebo in the subgroup of patients with blood eosinophil counts of 400 cells per μL or more (0.64, 95% CI 0.43–0.95). Greater reductions in annual exacerbation risk ($p=0.0035$) and longer time to first exacerbation were observed for patients with higher trough serum reslizumab concentrations. In study 2, we found no difference between placebo and fixed-dose subcutaneous reslizumab in categorised percentage reduction in daily oral corticosteroid dose (odds ratio for a lower category of oral corticosteroid use in the reslizumab group vs the placebo group, 1.23, 95% CI 0.70–2.16; $p=0.47$). The frequency of adverse events and serious adverse events with reslizumab were similar to those with placebo in both studies.

Interpretation Fixed-dose (110 mg) subcutaneous reslizumab was not effective in reducing exacerbation frequency in patients with uncontrolled asthma and increased blood eosinophils (≥ 300 cells/ μL), or in reducing the daily maintenance oral corticosteroid dose in patients with oral corticosteroid-dependent severe eosinophilic asthma. Higher exposures than those observed with 110 mg subcutaneous reslizumab are required to achieve maximal efficacy.

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Introduction

Eosinophils have a key role in the promotion and maintenance of airway inflammation, airway wall

thickening, fibrosis, and angiogenesis in patients with eosinophilic asthma.¹ Baseline counts of blood and sputum eosinophils can be markers of the severity of

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Research in context

Evidence before this study

We searched PubMed for publications from inception to May 3, 2019, using the terms “anti-interleukin-5 antibodies” or “reslizumab” and “asthma” for clinical trials investigating anti-interleukin 5 (IL5) biologic drugs in patients with severe eosinophilic asthma. In two phase 3 trials, weight-based dosing of intravenous reslizumab (3 mg/kg) reduced clinical asthma exacerbation frequency and improved asthma control, lung function, and quality of life. In other reslizumab studies, improvement in lung function was found with intravenous reslizumab (0.3 mg/kg and 3 mg/kg); the greatest improvement in lung function with the higher dose was observed in patients with a blood eosinophil count of 400 cells per μL or more. Clinical trials of other drugs that target the IL5 pathway were also identified. Studies have shown that intravenous or subcutaneous anti-IL5 or anti-IL5 receptor drugs reduce asthma exacerbations and improve lung function, asthma control, and quality of life, and reduce maintenance oral corticosteroid dose in patients with severe eosinophilic asthma.

Added value of this study

To our knowledge, these are the first published studies that have assessed the safety and efficacy of low-dose reslizumab administered as a fixed-dose subcutaneous injection for the treatment of severe eosinophilic asthma. The results of these studies suggest that the lower dose and lower systemic reslizumab concentrations associated with fixed-dose subcutaneous reslizumab, compared with intravenous reslizumab, might not be sufficient to provide robust reductions in exacerbation frequency in patients with uncontrolled asthma

and increased blood eosinophil counts (≥ 300 cells/ μL), or to reduce daily maintenance oral corticosteroid dose in patients with oral corticosteroid-dependent severe asthma. Results indicate that the exposure–response relationships and systemic reslizumab concentration thresholds differ between the endpoints of asthma exacerbations and lung function. Greater effect on asthma exacerbations was observed in the prespecified subgroup of patients with a high blood eosinophil count (≥ 400 cells/ μL), which matched the population in the phase 3 studies of intravenous reslizumab. Regional differences in background asthma exacerbation rate during the study period were also observed.

Implications of all the available evidence

A complete analysis of the effect of body weight on systemic concentrations and a clear understanding of the exposure–response relationships for primary and key secondary asthma endpoints, before initiation of phase 3 studies, would be beneficial to support optimal future study design for biologic drugs in severe asthma. Our results support further assessment of higher doses of weight-based subcutaneous reslizumab. Baseline blood eosinophil counts are a marker of patients who are more responsive to therapies targeting the IL5 pathway, reconfirming the importance of defining the eosinophilic phenotype; however, treatment-induced suppression of blood eosinophils with anti-IL5 therapy is not sufficient to ensure robust clinical efficacy. A reduction in blood eosinophil counts with treatment is a surrogate biomarker, which might not fully reflect changes in eosinophilic airway inflammation in patients with severe asthma. Regional differences in asthma exacerbation rates have been observed and should be considered during future study design.

asthmatic symptoms and independent risk factors for future asthma exacerbations in patients with symptomatic asthma.^{2–4}

Maintenance and activation of eosinophils is dependent on interleukin-5 (IL5), the prototypical eosinophil viability-enhancing factor.⁵ Several monoclonal antibodies that target the IL5 pathway are approved as add-on maintenance treatment for inadequately controlled severe eosinophilic asthma.^{6–8} Reslizumab is an IgG4 κ humanised anti-IL5 monoclonal antibody that is administered as intravenous infusion at a recommended weight-based dose of 3 mg/kg, once every 4 weeks.⁸ Efficacy and safety of intravenous reslizumab have been shown in five phase 3 clinical trials.^{9–12} Weight-based dosing of intravenous reslizumab significantly reduced the risk of asthma exacerbations and improved asthma control, lung function, and quality of life in patients with inadequately controlled asthma, peripheral blood eosinophil counts of 400 cells per μL or more, and a history of asthma exacerbations.⁹ The availability of subcutaneous administration as an alternative to intravenous administration could increase the accessibility of reslizumab treatment, reduce time and

resource requirements associated with intravenous infusion,^{13,14} and simplify administration.

We did two phase 3 clinical trials, which were designed to investigate the safety and efficacy of reslizumab given as a fixed-dose (110 mg) subcutaneous injection once every 4 weeks in patients with uncontrolled asthma and increased blood eosinophils. Study 1 was an asthma exacerbation study that investigated the effect of fixed-dose subcutaneous reslizumab on the rate of clinical asthma exacerbations, and study 2 was a corticosteroid-sparing study that investigated the potential oral corticosteroid-sparing properties of fixed-dose subcutaneous reslizumab in patients with oral corticosteroid-dependent severe asthma.

Methods

Study design

Both studies were randomised, double-blind, placebo-controlled, parallel-group, phase 3 studies. Full details of the study methods are provided in the appendix (p 3). Study 1 consisted of a 2-week screening period, a minimum 3-week run-in period, a 52-week treatment

See Online for appendix

period, and a 12-week follow-up. Study 2 had a 2-week screening period, followed by an oral corticosteroid dose optimisation period of up to 10 weeks, a minimum 2-week run-in period, a 24-week treatment period, and an 8-week follow-up. For study 1, at least one patient was randomly assigned at each of 155 study sites in 20 countries: USA (43 sites), Ukraine (19), Hungary (16), Germany (11), Argentina (10), Russia (8), Israel (8), Poland (6), Romania (5), South Africa (5), Japan (4), Mexico (3), Belgium (3), Czech Republic (3), Turkey (3), Canada (2), Spain (2), New Zealand (2), France (1), and Australia (1). For study 2, at least one patient was randomly assigned at each of 78 study sites in 17 countries: Ukraine (15), the USA (9), Germany (7), Poland (7), Argentina (6), Russia (6), Mexico (5), Israel (5), Spain (3), Australia (2), South Korea (2), France (2), the Netherlands (2), Belgium (2), Hungary (2), Czech Republic (2), and Italy (1).

The subcutaneous reslizumab fixed dose of 110 mg was selected on the basis of clinical data from an intravenous reslizumab phase 3, 16-week, FEV₁ study in patients with eosinophilic asthma,¹¹ in which improvements in lung function and asthma control and reductions in blood eosinophil counts were obtained with intravenous doses of 0·3 mg/kg and 3 mg/kg. Although data on asthma exacerbations were not collected during the treatment period, the similar magnitude of effect on FEV₁ between the 0·3 mg/kg and 3 mg/kg doses suggested that the doses would also have similar effects on asthma exacerbation frequency. A 110 mg subcutaneous dose approximates to a 1 mg/kg intravenous dose for the average 70 kg patient, when adjusting for subcutaneous bioavailability, and was therefore expected to result in clinical efficacy in most patients. The 1 mL volume and presentation in a pre-filled syringe would be convenient for subcutaneous administration.

Both studies were done in accordance with the Declaration of Helsinki and the International Conference on Harmonisation guidelines for Good Clinical Practice. All study documents and procedures were approved by the appropriate Institutional Review Board or Ethics Committee at each study site. All patients provided written informed consent before study participation.

Study 1

Participants

We included participants aged 12 years or older with a diagnosis of asthma, an Asthma Control Questionnaire-6 (ACQ-6) score of at least 1·5 at screening, two or more asthma exacerbations requiring the use of systemic corticosteroids in the past year, a blood eosinophil count of 300 cells per μ L or more during screening, and an FEV₁ reversibility of 12% or more to inhaled short-acting β agonist at screening or documentation within 12 months of signing informed consent. Patients were receiving at least a medium dose (>250 μ g/day fluticasone or equivalent) of inhaled corticosteroids for at least 3 months with at least one additional asthma controller

for at least 3 months (or documented failure of a second controller in the past 12 months). For patients using inhaled corticosteroid and long-acting β agonist (LABA) combinations, the mid-strength approved maintenance dose in the local country met this inhaled corticosteroid criterion. Oral corticosteroid use (prednisone 10 mg or less daily, or equivalent) was allowed. Full inclusion and exclusion criteria are summarised in the appendix (p 3).

Randomisation and masking

Patients were randomly assigned (1:1) to receive subcutaneous reslizumab (110 mg) or placebo. Participants were assigned using an interactive web response system (IWRS). Randomisation was stratified by age (12 to <18 years and $\geq$ 18 years) and blood eosinophil count at screening (300 to <400 cells/ μ L and $\geq$ 400 cells/ μ L). A maximum of 30% of patients enrolled could have a blood eosinophil count of between 300 cells per μ L and less than 400 cells per μ L; when this threshold was met, only patients with blood eosinophil counts of 400 cells per μ L or more were enrolled. The inclusion criterion of a baseline blood eosinophil count of 300 cells per μ L or more was based on the available published literature on biologic drugs for treating severe asthma at the time of study design.^{15–17} Anti-IL5 therapies have been shown to reduce asthma exacerbations in patients with baseline blood eosinophil counts of 150 cells per μ L or more¹⁵ or 300 cells per μ L or more.^{16,17} However, data for 3 mg/kg intravenous reslizumab suggested robust effects across multiple asthma endpoints in patients with baseline blood eosinophil counts of 400 cells per μ L or more; therefore, we restricted the enrolment of patients with blood eosinophils between 300 cells per μ L and less than 400 cells per μ L to 30%. Patients and investigators were masked to treatment assignment. The sponsor's personnel involved in the study were also masked to the study drug identity, until the database was locked for analysis and the treatment assignment was revealed.

Procedures

Eligible patients received either one 1·0 mL subcutaneous injection containing 110 mg reslizumab, or one 1·0 mL subcutaneous injection containing sterile placebo solution, once every 4 weeks. Reslizumab and placebo were provided as matching, clear solutions. Treatment continued for 52 weeks, with the last dose administered at week 48 followed by an end-of-treatment visit at week 52. Patients maintained their inhaled asthma controller regimen without change throughout screening, optimisation, run-in, treatment, and follow-up periods (appendix p 3).

Outcomes

The primary efficacy outcome was the frequency of clinical asthma exacerbations per patient during the 52-week treatment period. A clinical asthma exacerbation was defined as a clinically judged deterioration in asthma

	Asthma exacerbation study (study 1)		Oral corticosteroid-sparing study (study 2)	
	Placebo (n=230)*	Subcutaneous reslizumab (n=234)*	Placebo (n=89)	Subcutaneous reslizumab (n=88)
Age, years	44.8 (17.7)	46.9 (17.6)	53.1 (12.0)	55.5 (12.7)
Sex, n (%)				
Men	102 (44%)	90 (38%)	32 (36%)	28 (32%)
Women	128 (56%)	144 (62%)	57 (64%)	60 (68%)
Weight, kg				
Mean (SD)	76.9 (18.3)	78.3 (19.4)	82.7 (18.9)	79.6 (21.4)
Median (IQR)	75.0 (23.6)	77.5 (25.2)	82.0 (20.7)	75.6 (26.2)
BMI, kg/m ²	27.4 (6.1)	28.3 (6.2)	29.9 (6.3)	29.4 (8.0)
FEV ₁ reversibility, %	24.6% (21.3)	22.8% (17.1)	24.6% (20.5)	26.0% (25.9)
Pre-bronchodilator FEV ₁ , L	2.1 (0.9)	2.0 (0.8)	1.7 (0.7)	1.6 (0.7)
FEV ₁ , % predicted	65.0% (19.7)	64.4% (18.3)	58.7% (19.8)	55.2% (16.7)
ACQ-6 score	2.6 (0.8)	2.6 (0.8)	2.4 (1.2)	2.3 (1.1)
AQLQ score	4.4 (1.0)	4.3 (1.0)	4.4 (1.2)	4.4 (1.1)
Blood eosinophil count, cells/μL†	737 (484)	787 (538)	521 (491)	479 (388)
Eosinophil count				
<300 cells/μL	0	1 (<1%)	33 (37%)	28 (32%)
≥300 to <400 cells/μL	42 (18%)	48 (21%)	13 (15%)	19 (22%)
≥400 cells/μL	188 (82%)	185 (79%)	43 (48%)	41 (47%)
Phadiatop test				
Positive	154 (67%)	159 (68%)	47 (53%)	47 (53%)
Missing	12 (5%)	4 (2%)	3 (3%)	1 (1%)
Former smoker	28 (12%)	31 (13%)	13 (15%)	12 (14%)
Age at asthma onset, years	27.5 (19.2)	29.0 (19.9)	32.9 (17.9)	32.8 (17.7)
Time since diagnosis, years	17.5 (13.5)	18.0 (14.0)	20.1 (14.2)	22.3 (16.0)
Asthma exacerbations requiring systemic corticosteroids in the past 12 months	2.3 (0.8)	2.4 (1.0)	1.85 (1.2)	2.1 (1.6)
Inhaled corticosteroid	229 (>99%)‡	232 (>99%)‡	89 (100%)	88 (100%)
High-dose§	98 (43%)	99 (42%)	70 (79%)¶	70 (80%)¶
Medium-dose§	122 (53%)	128 (55%)	17 (19%)	18 (20%)
Long-acting β agonist	218 (95%)	224 (96%)	84 (94%)	87 (99%)
Oral corticosteroid	15 (7%)	15 (6%)	89 (100%)	88 (100%)
Aspirin sensitivity	23 (10%)	26 (11%)	7 (8%)	5 (6%)
Chronic rhinosinusitis with nasal polyps	38 (17%)	37 (16%)	13 (15%)	10 (11%)

Data are mean (SD), or n (%), unless otherwise indicated. ACQ-6=Asthma Control Questionnaire-6. AQLQ=Asthma Quality of Life Questionnaire. BMI=body-mass index. ITT=intention-to-treat. *Two patients in the placebo group and two patients in the reslizumab group were excluded from the ITT analysis. One site was terminated because of recurrent irregularities in data collection and recording. The patients who were randomly assigned at this site were excluded from the ITT analysis set. †Stated baseline eosinophil count reflected the highest value from measurements during screening, run-in, and pre-dose day of randomisation. ‡Three patients presented as not using inhaled corticosteroids at baseline; two (one in each treatment group) were using inhaled corticosteroids at baseline but were counted as non-inhaled corticosteroid users. Of these two patients, the first was using budesonide with formoterol fumarate, which was not coded as an inhaled corticosteroid, and the second was using beclomethasone with formoterol, which was not counted as a baseline medication because of missing dates (the dates were confirmed to be unknown, but were >3 months before screening). The third patient (in the reslizumab group) was not using inhaled corticosteroid at baseline. §High-dose was defined as >500 μg/day fluticasone or equivalent and medium-dose as >250 to ≤500 μg/day fluticasone or equivalent. ¶In regions where the inhaled corticosteroid dose in the highest labelled fixed-dose preparation of inhaled corticosteroid and long-acting β agonist was <500 μg/day fluticasone or equivalent, patients receiving the highest labelled dose in that region were deemed to satisfy the inclusion criterion. ||Self-reported.

Table 1: Demographics and baseline clinical characteristics of the intention-to-treat population in studies 1 and 2

control resulting in at least one of the following: use of systemic corticosteroids (oral or injection), at least a doubling from a stable maintenance oral corticosteroid dose for 3 days or more, or an asthma-specific hospital admission or emergency department visit. A pre-specified, powered subgroup analysis assessed the frequency of clinical asthma exacerbations per patient during the 52-week treatment period in patients who had baseline blood eosinophil counts of 400 cells per μL or more. Secondary outcomes included lung function, as assessed by measurement of change from baseline in pre-bronchodilator FEV₁; patient-reported outcomes (ie, change in ACQ-6 and AQLQ+12 scores at 52 weeks), safety; and immunogenicity (appendix p 13). Blood eosinophil counts at specified timepoints during the treatment period and follow-up visit were assessed (at 12 weeks after end of treatment); the highest value from screening to predose on the day of randomisation was used as baseline.

Serum reslizumab concentrations were determined from blood samples collected at baseline; before study drug administration at weeks 4, 8, 12, 16, 20, 32, 48, and 52; and at early withdrawal or at the follow-up visit. An additional sample was taken at long-term follow-up (appendix p 5).

Study 2

Participants

We included patients aged 12 years or older with a diagnosis of asthma, an FEV₁ reversibility of 12% or more after inhaled reliever medication at screening, or a historical FEV₁ reversibility or a positive methacholine challenge within 24 months, a blood eosinophil count of 300 cells per μL or more within the past 12 months while on at least a medium total daily dose of inhaled corticosteroids, during screening, during oral corticosteroid optimisation, or at the week -2 visit (appendix p 7). Patients required an average daily maintenance dose of oral corticosteroid for asthma (prednisone 5–40 mg, or equivalent) during the 3 months before screening. Patients also required high-dose inhaled corticosteroids (≥880 μg of inhaled fluticasone propionate, or equivalent) plus another controller for at least 6 months before screening. For a fixed-dose preparation of inhaled corticosteroid plus LABA, the highest labelled dose in the local country satisfied this criterion. For patients aged 12 to less than 18 years, at least a medium-dose inhaled corticosteroid plus another controller was required. The dose and regimen needed to have been stable for at least 30 days. Full inclusion and exclusion criteria are summarised in the appendix (p 8).

Randomisation and masking

Patients were randomly assigned (1:1) to receive subcutaneous reslizumab (110 mg) or placebo. The randomisation list and treatment were assigned to the relevant treatment groups using an IWRS. Randomisation

was stratified by optimised average daily oral corticosteroid dose (>10 mg or ≤ 10 mg) and age (12 to <18 years and ≥ 18 years) at baseline. Patients and investigators were masked to treatment assignment. The sponsor's personnel involved in the study were also masked to the study drug identity, until the database was locked for analysis and the treatment assignment was revealed.

Procedures

To achieve the patient's minimal effective oral corticosteroid requirement (appendix p 9), the oral corticosteroid dose was reduced at weekly intervals for up to 10 weeks or until asthma signs and symptoms worsened (minimum 1 day in optimisation). Oral corticosteroid optimisation was considered finished when patients had worsening of asthma signs and symptoms, or if patients optimised to an oral corticosteroid dose of less than 5 mg without worsening of asthma signs and symptoms.

Patients whose minimal effective oral corticosteroid dose remained in the range 5–40 mg prednisone daily at the end of optimisation could advance to run-in, after which patients received either one 1.0 mL subcutaneous injection containing 110 mg reslizumab, or matched placebo, once every 4 weeks. Treatment continued for 24 weeks, with the last dose administered at week 20, followed by an end-of-treatment visit at week 24. Patients maintained an inhaled asthma controller regimen without change throughout the study. The treatment period consisted of a stable-dose induction period of 4 weeks, followed by an oral corticosteroid reduction period. During oral corticosteroid reduction, the oral corticosteroid dose was reduced in accordance with the protocol, based on meeting asthma stability criteria at scheduled monthly clinic visits during weeks 5–20 (appendix p 9).

Outcomes

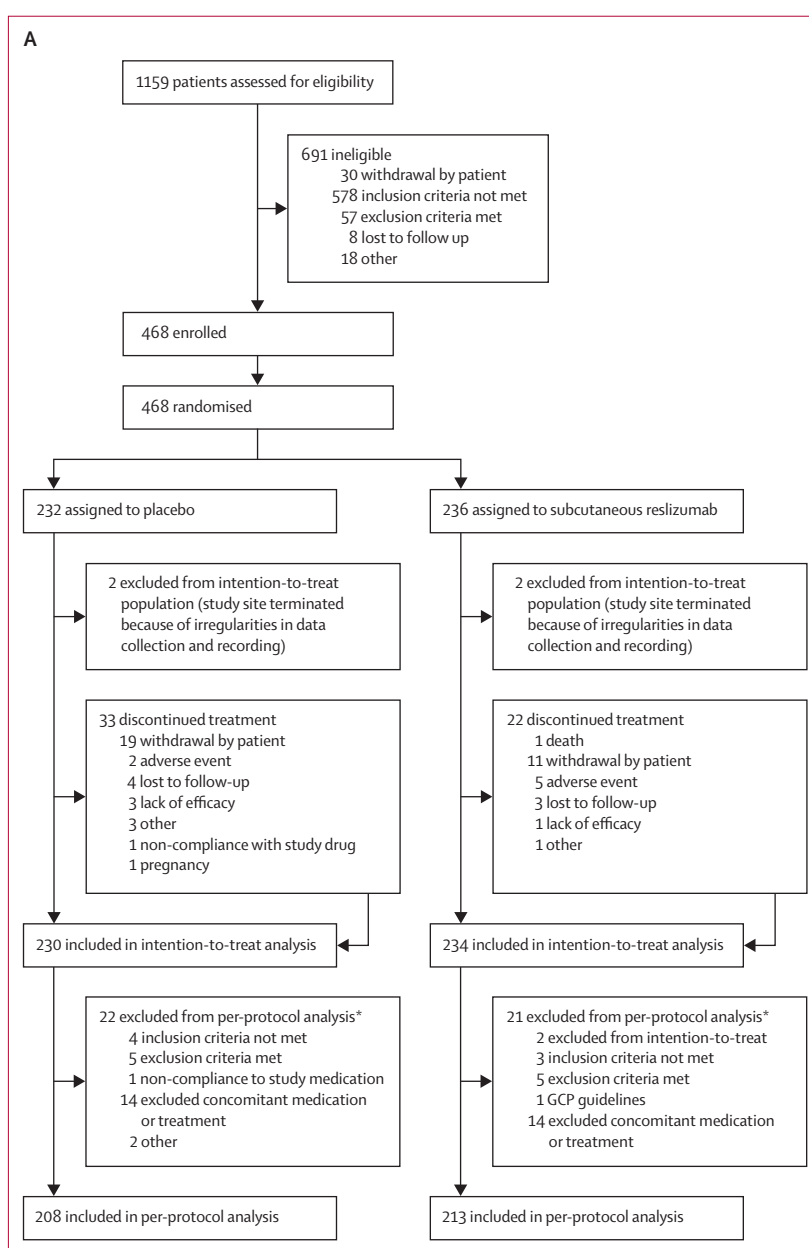
The primary efficacy outcome was categorised percentage reduction in daily oral corticosteroid dose compared with baseline (end of the optimisation phase) during weeks 20–24: 90 to 100%; 75 to $<90\%$; 50 to $<75\%$; >0 to $<50\%$; no decrease in oral corticosteroid dose or loss of asthma control during weeks 20–24; or treatment discontinuation (withdrawal of treatment). Loss of asthma control was defined as FEV₁ less than 80% of baseline at week 24, clinically significant worsening in ACQ-6 score (increase in score ≥ 0.5 points) at week 24 compared with baseline, or clinical asthma exacerbation during weeks 20–24. Patients with missing data or loss of asthma control with reduced oral corticosteroid dose were included in the category reflecting the lowest efficacy.

Secondary outcomes included additional assessments of oral corticosteroid dose reduction, clinical asthma exacerbations, safety, and immunogenicity (appendix p 10). The assessment of blood eosinophil counts was similar to that in study 1; the highest value from screening to week -2 was used as baseline.

Serum reslizumab concentrations were determined from blood samples collected at baseline; before study drug administration at weeks 4, 8, 12, 16, 20, and 24; and at early withdrawal or at the follow-up visit. An additional sample was taken at long-term follow-up (appendix p 7).

Statistical analysis

The primary efficacy analyses used the intention-to-treat (ITT) population of each study, which included all randomly assigned patients. Analyses were also done in the per-protocol populations, defined as all patients from the ITT population without major protocol violations.



(Figure 1 continues on next page)

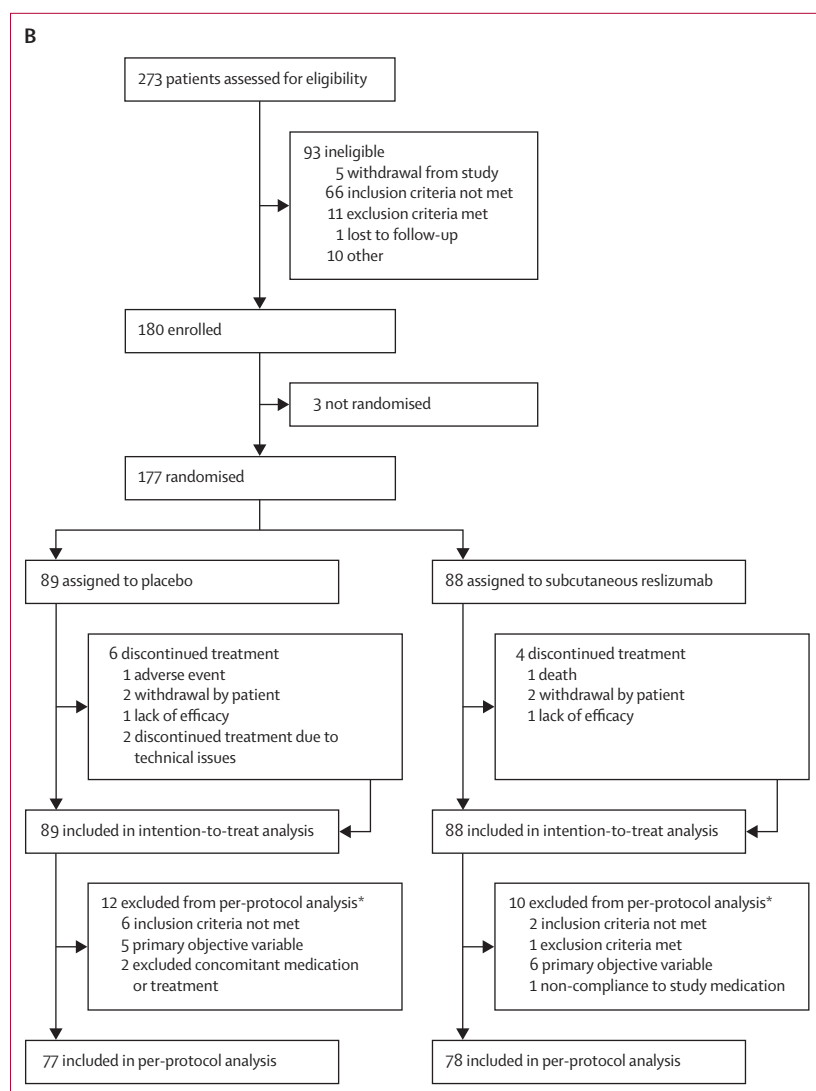


Figure 1: Trial profiles for studies 1 and 2

GCP=good clinical practice. *A patient might have had multiple reasons for exclusion from the per-protocol analysis.

Safety was assessed in the safety population of each study, which included all patients who received at least one dose of study treatment.

Simulation studies were done with SAS (version 9.4) to assess the statistical powers for analysing the frequency of exacerbations using a negative binomial model in study 1, and the power analysis software PASS (version 13.0.7) was used for sample size estimation in study 2. The model assumptions involved in the calculations are described in detail in the appendix (p 11). In brief, power calculations for study 1 were based on a mean of 2.9 exacerbations per patient per year in the placebo group, a dispersion parameter of 1.2 in the negative Binomial model for both treatment groups, and treatment effect of 45% between reslizumab and placebo. On the basis of these assumptions, 225 patients per group would provide more

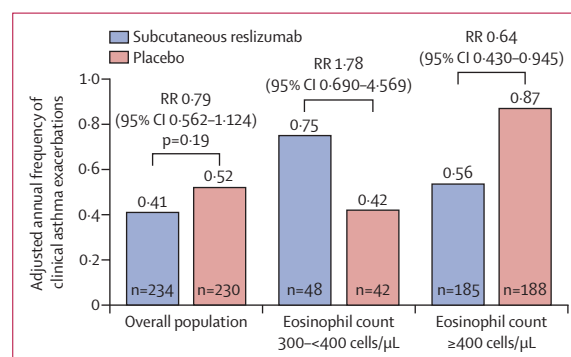


Figure 2: Adjusted frequency of clinical asthma exacerbations during 52 weeks in the intention-to-treat population and in subgroups of patients based on baseline eosinophil counts in study 1

For this analysis, the offset variable is calculated as the logarithm of treatment duration minus the summed duration of exacerbations during the treatment period. RR=rate ratio.

than 90% power to detect a significant treatment effect of reslizumab versus placebo in the reduction of exacerbation rate. For study 2, a sample size of 76 patients per treatment group was chosen to provide 90% power to detect a significant effect of reslizumab over placebo on the probability for a higher categorical reduction of oral corticosteroid dose using a proportional odds model. This was based on the assumption in the placebo group that after 24 weeks of treatment 10.9% of patients have 90–100% reduction, 7.9% have 75 to <90% reduction, 14.8% have 50 to <75% reduction, 10.8% have 0 to <50% reduction, and 55.6% have no reduction, have loss of asthma control, or discontinue; also assuming odds ratio (OR) based on a proportional odds model of 2.63.

For both studies, a fixed-sequence, multiple-testing procedure was implemented to test the primary and secondary outcomes while controlling the overall type I error rate at 0.05. Primary and secondary outcomes were tested sequentially and hierarchically. If the resulting two-sided p value from the primary comparison was less than 0.05, then the next comparison was interpreted inferentially. No multiplicity adjustments were made for other efficacy and exploratory analyses. All statistical analyses were done using SAS (versions 9.1.3 and 9.2).

The frequency of clinical asthma exacerbations was analysed with a generalised linear model for data from the negative binomial regression model. Treatment group, randomisation stratification factors (age and baseline blood eosinophil count for study 1; age and oral corticosteroid dose for study 2), and number of exacerbations in the previous year were model factors. Logarithm of follow-up duration minus the summed duration of exacerbations was an offset variable. Adjusted clinical asthma exacerbation rates, rate ratios (RR), and 95% CIs were calculated. In study 1, multiple imputation was used for missing data in the primary analysis model. Sensitivity analyses were done without imputed data. Because of convergence issues with the primary analysis

model in study 1, a post-hoc data-driven analysis of the effects of subcutaneous reslizumab was done (appendix p 6).

Change from baseline in pre-bronchodilator FEV₁ was analysed with a mixed-model repeated measures model, with fixed effects for treatment, visit, treatment-by-visit interaction, sex, age group, and blood eosinophil group; height and baseline value as covariates; and patient as a random effect. Unstructured covariance was assumed for the repeated measures.

In study 2, categorised percentage reduction in total daily oral corticosteroid dose at weeks 20–24 was analysed with a proportional odds model including factors for treatment group and randomisation strata (age group and oral corticosteroid dose). Baseline oral corticosteroid dose and duration of oral corticosteroid use before study entry were also included as covariates. ORs, 95% CIs, and p values were calculated. A prespecified sensitivity analysis was done with the per-protocol population.

Statistical methods for the secondary efficacy and safety outcomes in both studies are provided in the appendix (pp 6, 11). Prespecified subgroups for the subgroup analyses are detailed in the appendix (pp 7, 12). Model results were only reported if there were 15 patients or more contributing to the analysis in each treatment group; otherwise, descriptive statistics were generated. Eosinophil count (300 to <400 cells/ μ L and $\geq$ 400 cells/ μ L) at baseline was a prespecified stratification factor for study 1, which recruited a sufficient proportion of patients with eosinophil counts of 400 cells per μ L or more (a minimum of 70% of the study population, per protocol). This subgroup analysis was prespecified for the primary analysis and selected secondary efficacy analyses.

The exposure–response analysis was assessed a posteriori using SAS (version 9.4). We identified the relationship between individual reslizumab trough concentrations (C_{trough}) and clinical asthma exacerbation

rate in study 1 using visual graphical analysis with a regression coefficient. Individual C_{trough} was based on the aggregate information for each individual using a model that accounted for the differences in time after the most recent dose and provided a single concentration measure for use in the exposure–response analysis. Patients in the placebo group were assumed to have concentrations of 0 μ g/mL. The underlying hazard for a clinical asthma exacerbation was explored with Kaplan-Meier survival curves of the time to first clinical asthma exacerbation, stratified by tertiles of reslizumab C_{trough} in study 1.

These studies are registered with ClinicalTrials.gov, NCT02452190 (study 1) and NCT02501629 (study 2).

Role of the funding source

Three authors of the report (DAM, MG, and RV) are employees of Teva Branded Pharmaceutical Products R&D, the study funder, and two (JM and LH) are former employees of the study funder. Employees of the study funder had roles in the design of the study, data collection, data analysis, data interpretation, and writing of the report. All authors had full access to all of the data outputs, and the corresponding author had final responsibility for the decision to submit for publication.

Results

Between Aug 12, 2015, and Oct 10, 2016, 1159 patients were recruited in study 1, of whom 468 patients were enrolled and randomly assigned to receive placebo (n=232 [50%]) or subcutaneous reslizumab (n=236 [50%]). The last visit of the last patient occurred on Jan 31, 2018. Between Sept 22, 2015, and Dec 12, 2016, 273 patients were recruited in study 2, of whom 180 were enrolled and 177 were randomly assigned to receive placebo (n=89) or subcutaneous reslizumab 110 mg (n=88). The last visit of the last patient occurred on Dec 4, 2017. Within each study, demographic and baseline disease characteristics were

	ITT population		Eosinophil count 300 to <400 cells/ μ L		Eosinophil count $\geq$ 400 cells/ μ L	
	Placebo (n=230)	Subcutaneous reslizumab (n=234)	Placebo (n=42)	Subcutaneous reslizumab (n=48)	Placebo (n=188)	Subcutaneous reslizumab (n=185)
Adjusted CAEs excluding exacerbation duration from the offset*	0.52	0.41	0.42	0.75	0.87	0.56
RR (95% CI) for reslizumab vs placebo	..	0.79 (0.56 to 1.12)	..	1.78 (0.69 to 4.57)	..	0.64 (0.43 to 0.95)
p value	..	0.19	..	..	..	..
Change in pre-bronchodilator FEV ₁ at 52 weeks, L, LSM (SE)†	0.23 (0.040)	0.37 (0.039)	0.25 (0.093)	0.38 (0.092)	0.26 (0.042)	0.41 (0.041)
Absolute increase (95% CI) vs placebo	..	0.14 (0.057 to 0.23)	..	0.13 (–0.053 to 0.31)	..	0.15 (0.054 to 0.25)
Change in ACQ-6 at 52 weeks, LSM (SE)‡	–1.14 (0.080)	–1.22 (0.078)	–1.36 (0.19)	–1.27 (0.19)	–1.05 (0.084)	–1.18 (0.081)
Absolute change (95% CI) vs placebo	..	–0.09 (–0.27 to 0.10)	..	0.09 (–0.31 to 0.48)	..	–0.12 (–0.32 to 0.083)
Change in AQLQ+12 at 52 weeks, LSM (SE)*§	1.06 (0.089)	1.14 (0.087)	1.25 (0.20)	1.30 (0.20)	1.02 (0.093)	1.11 (0.091)
Absolute increase (95% CI) vs placebo	..	0.08 (–0.11 to 0.27)	..	0.05 (–0.38 to 0.49)	..	0.08 (–0.14 to 0.30)

ACQ-6=Asthma Control Questionnaire-6. AQLQ+12=Asthma Quality of Life Questionnaire for patients aged $\geq$ 12 years. CAE=clinical asthma exacerbation. ITT=intention-to-treat. LSM=least squares mean. RR=rate ratio. *For this analysis, the offset variable was calculated as the logarithm of treatment duration minus the summed duration of exacerbations during the treatment period. †Minimal clinically important difference was a change of 0.1 L. ‡Minimal clinically important difference was change in score of –0.5. §Minimal clinically important difference was change in score of 0.5.

Table 2: Primary and key secondary efficacy outcomes in study 1

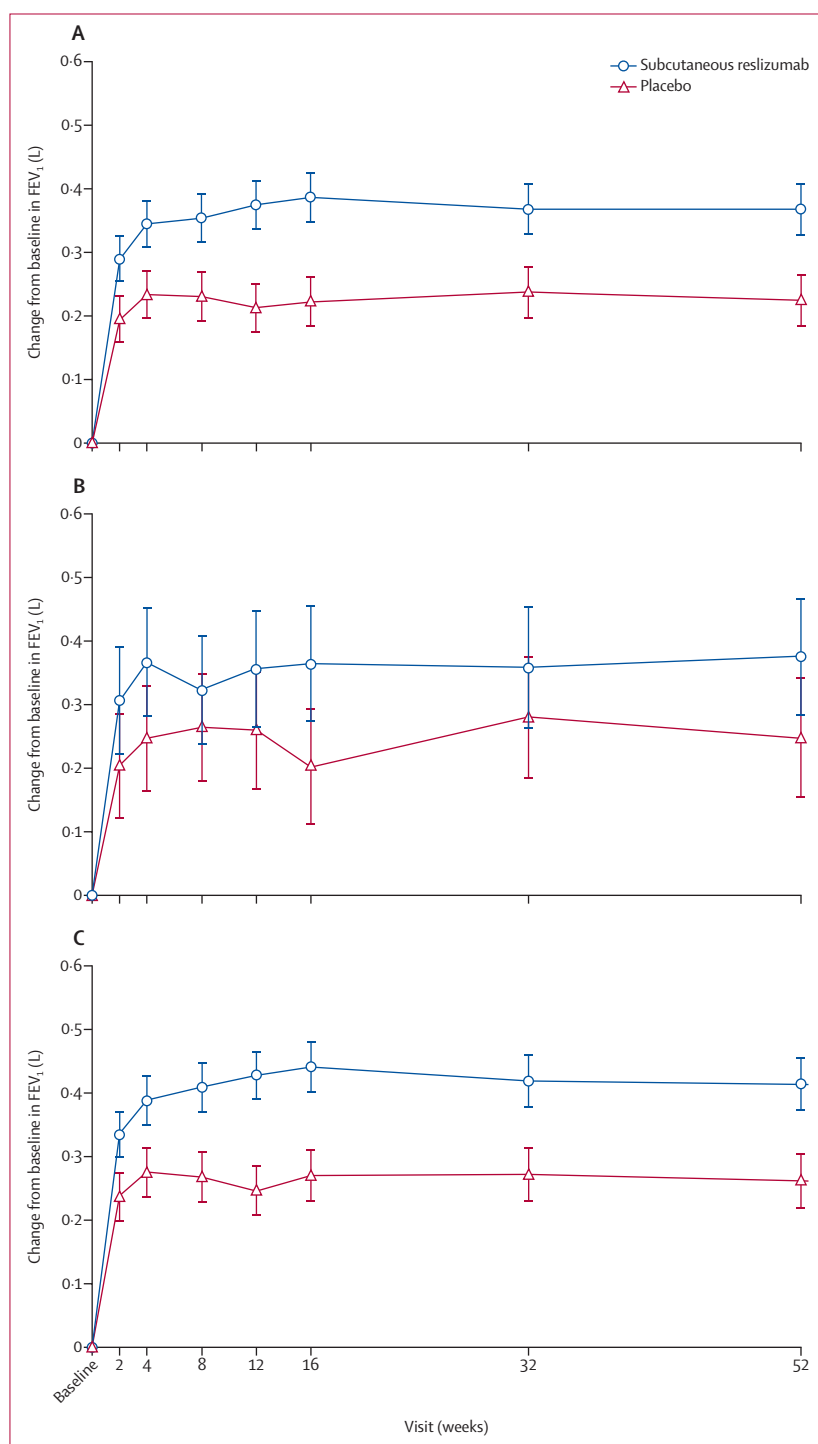


Figure 3: Change in pre-bronchodilator FEV₁ from baseline at each visit by treatment group in study 1
 (A) The intention-to-treat population, (B) patients with baseline eosinophil counts of 300 to <400 cells per µL, and (C) patients with baseline eosinophil counts of ≥400 cells per µL. Least squares mean and SE are given for the change from baseline in FEV₁ at each clinic visit between the first dose of study drug and the end-of-treatment visit for completed patients, and between the first dose of study drug and 4 weeks after the last dose of study drug for patients who discontinued treatment early. Inferential statistics are from a mixed-model repeated measures model with fixed effects for treatment, visit, treatment-by-visit interaction, sex, age group, and blood eosinophil group; height and baseline value as covariates; and patient as a random effect. Unstructured covariance was assumed for the repeated measures.

well balanced across the two treatment groups (table 1). The demographic and baseline disease characteristics for patients in study 1 by baseline eosinophil subgroup are shown in the appendix (p 13). Trial profiles are shown in figure 1. In study 1, 59 (13%) of 468 patients discontinued study treatment (35 [15%] of 232 in the placebo group and 24 [10%] of 236 in the reslizumab group). Seven patients were reported as lost to follow-up (four in the placebo group and three in the reslizumab group). In study 2, 10 (6%) of 177 patients discontinued study treatment (six [7%] of 89 in the placebo group and four [5%] of 88 in the reslizumab group).

In study 1, the frequency of clinical asthma exacerbations did not differ significantly over 52 weeks between fixed-dose subcutaneous reslizumab and placebo in the ITT population (RR 0.79, 95% CI 0.56–1.12; $p=0.19$; figure 2, table 2). As the analysis of the primary efficacy outcome did not meet the criteria for statistical significance, subsequent efficacy outcomes were not interpreted inferentially (figure 2, table 2). Fixed-dose subcutaneous reslizumab reduced clinical asthma exacerbation frequency compared with placebo in the prespecified subgroup of patients with eosinophil counts of 400 cells per µL or more (0.64, 95% CI 0.43–0.95), but not in those with eosinophil counts of between 300 cells per µL and less than 400 cells per µL (figure 2, table 2).

In the prespecified subgroups, patients enrolled in Europe had the smallest treatment effect, and adults aged 65 years or older had a slightly greater reduction in the frequency of clinical asthma exacerbations with fixed-dose subcutaneous reslizumab compared with placebo (appendix p 14). The findings of the other prespecified subgroup analyses were consistent with those of the primary analysis (appendix p 14). In a post-hoc analysis of patients enrolled in eastern Europe (110 [47%] in the reslizumab group and 99 [43%] in the placebo group), the treatment effect was smaller than for the overall results from all of Europe (data not shown), apparently driven by a low frequency of clinical asthma exacerbations in the placebo group (mean of 0.3 [standard error 0.06] exacerbations during the treatment period). Results in the per-protocol sensitivity analysis were similar to those in the primary ITT analysis (data not shown).

In the ITT population, the change from baseline in pre-bronchodilator FEV₁ at week 52 was larger with fixed-dose subcutaneous reslizumab compared with placebo (figure 3, table 2). Improvements in pre-bronchodilator FEV₁ with fixed-dose subcutaneous reslizumab were seen rapidly (ie, at 2 weeks) and were maintained throughout 52 weeks (figure 3). Increases from baseline at week 52 for pre-bronchodilator FEV₁ were larger for fixed-dose subcutaneous reslizumab than placebo in both the prespecified eosinophil subgroups (figure 3, table 2).

Overall ACQ-6 scores decreased from baseline to week 52 in both treatment groups in the ITT population; however, ACQ-6 scores were lower for fixed-dose

subcutaneous reslizumab at all postbaseline timepoints (appendix p 15). The least squares mean change in ACQ-6 score from baseline to week 52 did not differ for fixed-dose subcutaneous reslizumab and placebo (table 2, appendix p 15). ACQ-6 findings for the prespecified eosinophil subgroups are provided in table 2. Increases in Asthma Quality of Life Questionnaire for 12 years and older (AQLQ+12) were observed with both treatment groups. Changes from baseline in AQLQ+12 scores at 52 weeks were similar for subcutaneous reslizumab and placebo in the ITT population (table 2) and prespecified eosinophil subgroups (table 2). Treatment differences between fixed-dose subcutaneous reslizumab and placebo did not approach the minimal clinically important difference in the ITT population or in either of the prespecified eosinophil subgroups.

We found no meaningful difference between results of the other prespecified subgroup analyses (appendix p 7) and those of the ITT analysis for the key secondary outcomes (data not shown). Results of the other secondary efficacy analyses (Asthma symptom assessment at week 52 [based on a daily asthma control diary]; percentage of asthma control days at week 52 from baseline; change from baseline in St George's Respiratory Questionnaire score at week 32; probability of having a clinical asthma exacerbation by week 52; frequency of clinical asthma exacerbations requiring hospitalisation or emergency

department visit; and frequency of moderate asthma exacerbations, defined as exacerbations requiring additional asthma controlling medication that was not a systemic corticosteroid, and that did not result in an asthma-specific hospitalisation or emergency department visit during the 52-week treatment period) are presented in the appendix (p 7).

In study 2, we found no statistically significant difference between reslizumab and placebo across the five-level categorised percentage reductions in daily oral corticosteroid dose during weeks 20–24 compared with baseline (end of the optimisation phase; OR 1.23, 95% CI 0.70–2.16; $p=0.47$; table 3). Results in the per-protocol sensitivity analysis were similar to those of the primary ITT analysis (appendix p 19). No notable differences between fixed-dose subcutaneous reslizumab and placebo were observed for patients in any of the prespecified subgroup analyses (appendix p 19).

	Placebo (n=89)	Subcutaneous reslizumab (n=88)
Categorised percentage oral corticosteroid dose reduction (5-level response) at weeks 20–24*		
90 to 100%	20 (22%)	18 (20%)
75 to <90%	4 (4%)	8 (9%)
50 to <75%	8 (9%)	13 (15%)
>0 to <50%	9 (10%)	7 (8%)
No decrease†	48 (54%)	42 (48%)
Reslizumab vs placebo	..	1.23 (0.70–2.16)
p value	..	0.47
Categorised oral corticosteroid dose reduction (responder analyses) at weeks 20–24		
At least 50% reduction from baseline	32 (36%)	39 (44%)
Reslizumab vs placebo	..	1.45 (0.79–2.68)
Oral corticosteroid dose ≤5 mg	34 (38%)	37 (42%)
Reslizumab vs placebo	..	1.19 (0.63–2.23)
Oral corticosteroid dose 0 mg	20 (22%)	18 (20%)
Reslizumab vs placebo	..	0.82 (0.37–1.82)
At least 5 mg reduction from baseline	31 (35%)	36 (41%)
Reslizumab vs placebo	..	1.36 (0.72–2.56)

Data are n (%) or OR (95% CI), unless otherwise indicated. OR=odds ratio. *All data were included; missing data were included as non-responders. †No decrease in oral corticosteroid dose, loss of baseline asthma control during weeks 20–24, or discontinuation from study drug.

Table 3: Oral corticosteroid outcomes in the intention-to-treat population in study 2

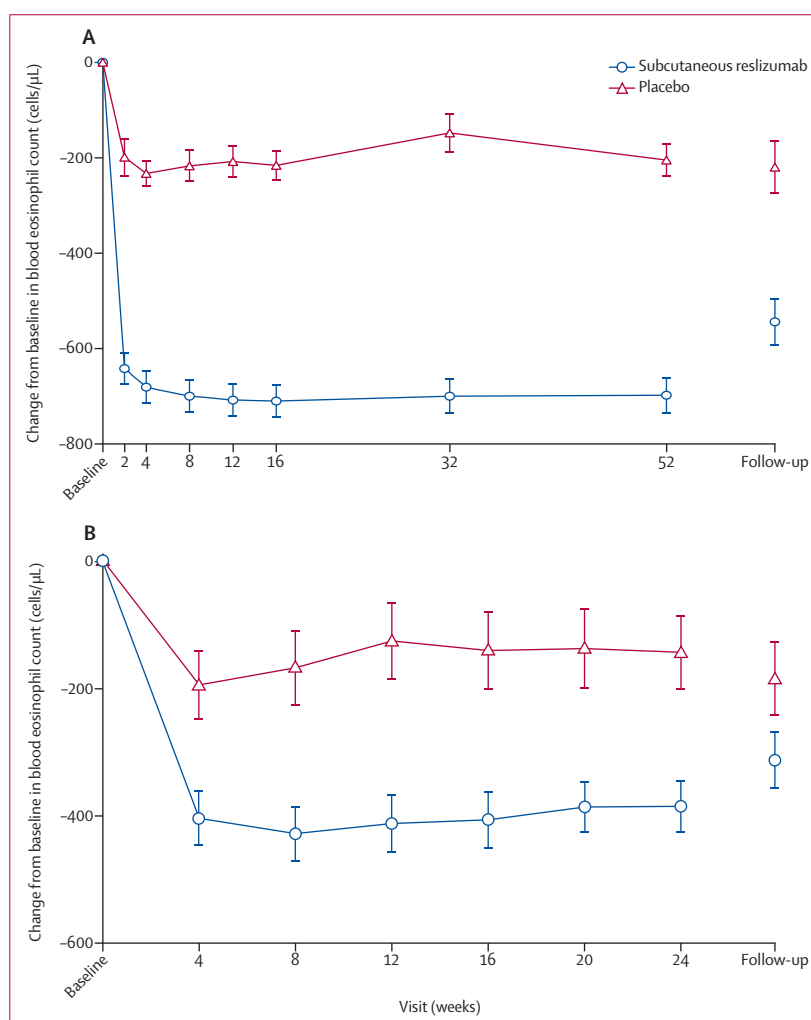


Figure 4: Change in blood eosinophil count from baseline at each visit in the intention-to-treat population in studies 1 and 2

Data are least squares mean and SE for (A) study 1 and (B) study 2. Follow-up was at 12 weeks after the 52-week treatment period in study 1, and at 8 weeks after the 24-week treatment period in study 2.

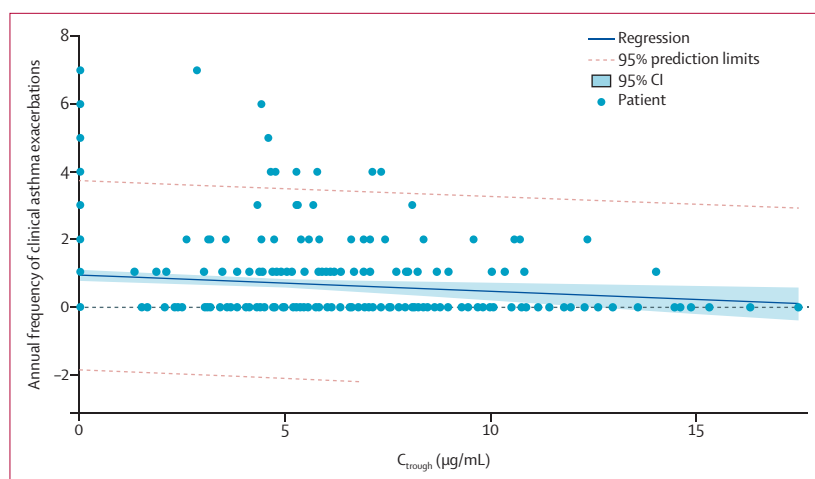


Figure 5: Relationship between trough serum concentration of reslizumab and annual frequency of clinical asthma exacerbations in study 1

C_{trough} =reslizumab trough concentration.

Because the analysis of the primary efficacy outcome did not meet the criteria for statistical significance, secondary efficacy outcomes were not interpreted inferentially. Both treatment groups showed decreases in oral corticosteroid dose over the treatment period (table 3). We found no significant differences in the secondary efficacy analyses for the ITT population (appendix p 20) or subgroup analyses of secondary outcomes (data not shown).

In both studies, blood eosinophil counts decreased in both treatment groups between baseline and first visit and remained below baseline until the end of treatment. Eosinophil counts began to rise towards baseline at the follow-up visits. Mean blood eosinophil counts among patients receiving fixed-dose subcutaneous reslizumab were lower than those of patients receiving placebo at the first post-baseline visit and all subsequent timepoints (figure 4). In study 1, the mean change from baseline to week 52 was $-0.205 \times 10^9/\text{L}$ (SD 0.4709) in the placebo group and $-0.698 \times 10^9/\text{L}$ (0.5356) in the fixed-dose subcutaneous reslizumab group. In study 2, the mean change from baseline to week 24 was $-0.144 \times 10^9/\text{L}$ (0.5040) in the placebo group and $-0.385 \times 10^9/\text{L}$ (0.3570) in the fixed-dose subcutaneous reslizumab group.

Baseline and reslizumab C_{trough} for both studies are reported in the appendix (pp 24–25). Patient body weight and C_{trough} were correlated and higher reslizumab C_{trough} was observed in patients with a lower body weight (appendix p 26).

In study 1, higher C_{trough} values were associated with a lower annual frequency of clinical asthma exacerbations. A regression fit indicated a significant correlation between clinical asthma exacerbation frequency and individual C_{trough} values ($p=0.0035$; figure 5).

The time to first clinical asthma exacerbation analysis indicated the greatest treatment effect (longest time to

first clinical asthma exacerbation) in patients with C_{trough} in the upper tertile (the group with the highest C_{trough} exposures). The C_{trough} exposures were $4.90 \mu\text{g/mL}$ or less for the lowest tertile, between 4.90 and less than $7.46 \mu\text{g/mL}$ for the middle tertile, and more than $7.46 \mu\text{g/mL}$ for the upper tertile, with the placebo group assumed to have an exposure of $0 \mu\text{g/mL}$. The median C_{trough} was $5.99 \mu\text{g/mL}$ (IQR 4.52 – 8.20 ; figure 6).

Within each study, the incidence of adverse events (AEs) was similar across the two treatment groups (table 4). The most common AEs ($\geq 5\%$ in either group) in study 1 were (viral or unspecified) upper respiratory tract infection, allergic rhinitis, acute sinusitis, headache, bronchitis, injection-site erythema, and asthma; and in study 2 were viral upper respiratory tract infection, asthma, bronchitis, headache, influenza, injection-site pain, and allergic rhinitis. Injection-site reactions occurred in 19 (8%) of 231 patients in the placebo group and 27 (11%) of 237 patients in the reslizumab group in study 1, and in two (2%) of 89 patients in the placebo group and five (6%) of 88 patients in the reslizumab group in study 2.

The most common serious AEs (SAEs) across both studies with reslizumab were asthma (table 4). In study 1, SAEs were assessed by investigators as being not related to study treatment, with the exception of two patients in the reslizumab group (respiratory infection and pulmonary embolism in one patient, and benign tracheal and bronchial tumour in another patient). In study 2, no SAEs during the study were assessed as being related to treatment.

One death occurred during the treatment period across both studies. In study 2, an adult man died 7 days after the first dose of reslizumab. The death was reported as sudden death and no autopsy was done. The death was considered by the investigator to be unrelated to treatment. In study 1, one patient from the reslizumab group died from hospital-acquired pneumonia and septicaemia during the follow-up period, more than 1 month after the last subcutaneous reslizumab dose. This event was not considered to be related to the study drug.

One ($<1\%$) patient in the placebo group and five (2%) patients in the fixed-dose subcutaneous reslizumab group discontinued treatment due to AEs in study 1. These AEs were of a diverse nature and, apart from benign tracheal and bronchial tumour and increased liver enzymes, were considered by the investigator to be not related to the study drug. In study 2, one patient in the placebo group discontinued treatment because of an AE.

No hypersensitivity AEs were deemed by the investigator to be causally related to reslizumab (appendix p 27).

Positive anti-drug antibody response in the reslizumab group was noted in 17 (7.2%) of 236 patients in study 1, and 11 (13%) of 88 patients in study 2. Titres of the anti-drug antibody-positive responses were generally low. All titres were transient in study 2 and most were

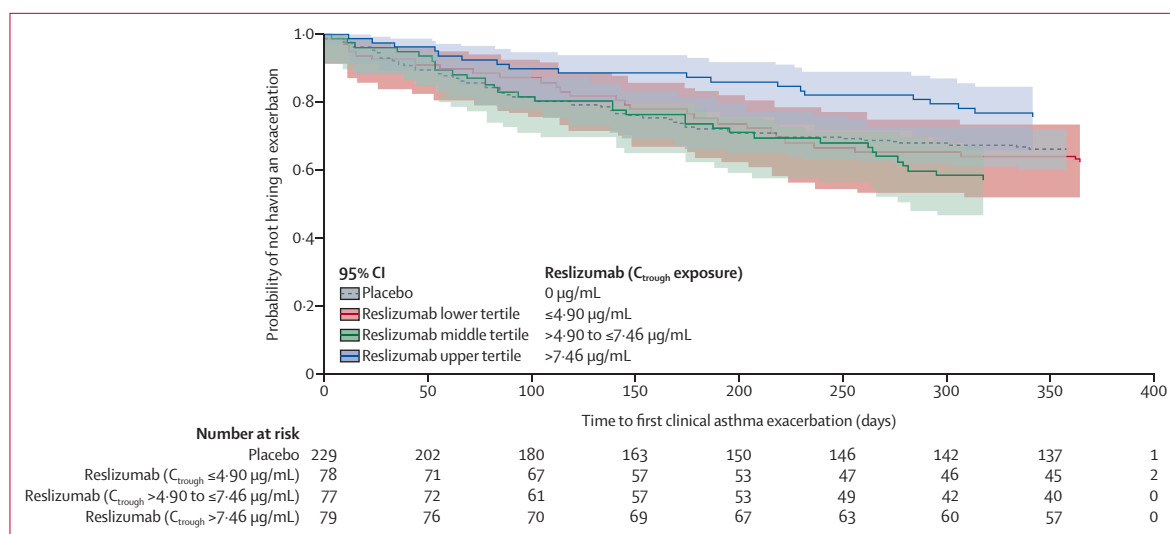


Figure 6: Kaplan-Meier survival curves for time to first clinical asthma exacerbation by tertiles of trough serum concentration of reslizumab in study 1. One patient in the placebo group received reslizumab and was excluded from the analysis. C_{trough} = trough reslizumab concentration.

transient in study 1 (most being single observations that resolved). We found no association between positive anti-drug antibody response and efficacy or safety outcomes (data not shown). All anti-drug antibody-positive samples were analysed using the neutralising antibody assay and none was identified as having neutralising activity.

Discussion

In study 1, 52 weeks of add-on therapy with fixed-dose (110 mg) subcutaneous reslizumab did not result in a significantly different frequency of clinical asthma exacerbations, compared with placebo, in the ITT population; however, our findings indicate that reslizumab treatment was associated with a reduced risk of exacerbations in the prespecified subgroup of patients with eosinophil counts of 400 cells per μL or more. This subgroup was fully powered a priori and aligned with the study population in phase 3 trials of intravenous reslizumab.⁹ The results of the secondary efficacy analyses suggest a clinically significant improvement in lung function, as measured by FEV_1 , which was greater with reslizumab compared with placebo at all timepoints. Improvements in lung function were already seen by the first measured timepoint, at week 2, and were sustained over 52 weeks. We observed no effect of subcutaneous reslizumab on the annual rate of clinical asthma exacerbations in the prespecified subgroup of patients with an eosinophil count between 300 cells per μL and less than 400 cells per μL , in whom eosinophilia is less likely to be the major contributor to asthma severity; however, the sample size was too small to draw conclusions for an event-driven endpoint such as asthma exacerbations, as indicated by the high variability in this subgroup. The finding of improvements in clinical asthma exacerbation frequency and lung function with reslizumab in the prespecified subgroup of patients with

	Study 1		Study 2	
	Placebo (n=231)	Subcutaneous reslizumab (n=237)	Placebo (n=89)	Subcutaneous reslizumab (n=88)
Any adverse event	172 (74%)	169 (71%)	47 (53%)	57 (65%)
Adverse events occurring in $\geq 5\%$ of patients by preferred term				
Upper respiratory tract infection	19 (8%)	28 (12%)	5 (6%)	1 (1%)
Viral upper respiratory tract infection	31 (13%)	27 (11%)	5 (6%)	11 (13%)
Allergic rhinitis	12 (5%)	16 (7%)	0	4 (5%)
Acute sinusitis	15 (6%)	14 (6%)	3 (3%)	2 (2%)
Headache	10 (4%)	14 (6%)	3 (3%)	4 (5%)
Bronchitis	17 (7%)	13 (5%)	4 (4%)	6 (7%)
Injection-site erythema	7 (3%)	13 (5%)	0	0
Injection-site pain	9 (4%)	10 (4%)	1 (1%)	4 (5%)
Asthma	21 (9%)	12 (5%)	6 (7%)	8 (9%)
Influenza	7 (3%)	10 (4%)	2 (2%)	4 (5%)
Hypertension	10 (4%)	6 (3%)	5 (6%)	1 (1%)
Any serious adverse events	19 (8%)	19 (8%)	4 (4%)	10 (11%)
Serious adverse events occurring in $\geq 2\%$ of patients by preferred term				
Asthma	5 (2%)	4 (2%)	2 (2%)	3 (3%)
Pneumonia	4 (2%)	2 (<1%)	0	1 (1%)
Cholecystolithiasis	4 (2%)	0	0	0
Adverse events leading to discontinuation from treatment	1 (<1%)	5 (2%)	1 (1%)	0
Deaths	0	0	0	1 (1%)

Data are n (%), based on the number of patients who had at least one adverse event of a particular classification. One patient in the placebo group received reslizumab.

Table 4: Adverse events during treatment by treatment group (safety population) in both studies

high blood eosinophil counts suggests that blood eosinophil counts are a marker of responsiveness to therapies targeting the IL5 pathway,^{15,16} reconfirming the importance of defining the eosinophilic phenotype. Indeed, the findings for reslizumab in the two

prespecified eosinophil subgroups suggest that a baseline threshold of 400 cells per μL or more is an indicator of the efficacy of reslizumab. These results are consistent with the findings of our intravenous reslizumab study,⁹ which showed no further reduction in the frequency of clinical asthma exacerbations with increasing baseline eosinophil counts higher than 400 cells per μL .

The primary outcome findings from study 1 for subcutaneous reslizumab differ from those of previous phase 3 studies of weight-based dosing of intravenous reslizumab,⁹ which showed a 50–59% reduction in clinical asthma exacerbations over 52 weeks, and these differences are probably at least partly due to differences in drug exposure between the two studies. The dose administered with subcutaneous reslizumab 110 mg approximates to a 1 mg/kg intravenous dose for the average 70 kg patient when accounting for bioavailability. Therefore, 110 mg administered subcutaneously was a substantially lower dose than that obtained with 3 mg/kg weight-based dosing for all observed body weights when accounting for bioavailability. For example, the median body weight in study 1 was 76 kg, and patients of this weight would have received a reslizumab dose of approximately 228 mg with 3 mg/kg weight-based dosing compared with the 110 mg subcutaneous dose used in these studies. The 110 mg subcutaneous dose was selected on the basis of previous clinical data for intravenous reslizumab at a lower dose of 0.3 mg/kg in a 16-week FEV₁ study,¹¹ and at the time the effect of exposures at lower doses on clinical asthma exacerbation rates was not as clearly understood. Additionally, for a monoclonal antibody with a soluble target such as IL5, for which clearance of the drug is influenced by body weight, a single fixed dose most likely results in a gradient of concentrations based on the distribution of body weights across a population. Indeed, our correlative data showed reduced drug exposure to fixed-dose reslizumab in patients with higher body weight. By contrast, weight-based dosing of intravenous reslizumab resulted in consistent exposures across body weights, thereby decreasing the risk of lower exposures in heavier patients.^{18,19}

Higher doses of intravenous reslizumab have previously been shown to be associated with larger reductions in blood eosinophil counts and greater improvements in lung function (FEV₁) and patient-reported outcomes,¹¹ compared with lower doses, supporting an exposure–response relationship for reslizumab. This relationship was confirmed in the exposure–response analysis in study 1. The exposures achieved in study 1 with fixed-dose subcutaneous reslizumab were sufficient for improvements in mean change from baseline in FEV₁, but exposures higher than those observed with 110 mg are presumably required to reach maximal efficacy in terms of reducing the exacerbation risk in most patients. Our exposure–response analysis for reductions in the annual frequency of clinical asthma exacerbations in study 1 showed an inverse relationship between exposure and

exacerbation rate, with higher reslizumab exposures associated with lower exacerbation frequencies, suggesting the need for higher exposures to reach the concentration threshold required for maximal efficacy in the greatest proportion of patients. Similar results were observed in the exposure–response analysis of time to first clinical asthma exacerbation.

The effect of body weight on other treatments targeting the IL5 pathway has also been assessed. Body weight was identified as a relevant covariate influencing the pharmacokinetics of other anti-IL5 drugs.^{20,21} The exposure–response relationship and the effect of patient body weight on the efficacy of treatments targeting the IL5 pathway should be explored in more detail in further studies or analyses.

In study 1, annual rates of clinical asthma exacerbations during the treatment period were low with placebo compared with the baseline period (12 months before screening). This finding might be partly due to treatment non-adherence or suboptimal treatment before study enrolment, which minimised treatment differences in the study. Additionally, results from our prespecified subgroup analyses in study 1 suggested that heterogeneity in regional exacerbation rates during the treatment period might have contributed to the size of the treatment effect: patients enrolled from eastern Europe and, to a lesser extent, South America, had a very low frequency of exacerbations during the treatment period, irrespective of treatment group. Similar findings were made in other studies of biologic drugs for the treatment of severe asthma,¹⁷ without clear cause, and additional studies are warranted. Regardless of these regional differences, the relationship between body weight and reslizumab concentrations and the consequent exposure–response relationship were evident across the study population.

Despite the evidence for pharmacological activity with reslizumab in study 1, we found no significant treatment differences in efficacy outcomes in study 2, which was designed to investigate the oral corticosteroid-sparing effect of reslizumab in patients who had more severe asthma at baseline than did participants in study 1. Patients in study 2 were older, had a longer duration of asthma, worse lung function, and higher body weight than did those in study 1. Because fewer low-weight patients were included in study 2 compared with study 1, reslizumab exposures were lower in study 2. Our findings suggest that the systemic drug concentrations obtained with reslizumab 110 mg were not sufficient to show a quantifiable difference between treatment groups in corticosteroid dose reduction in most patients with severe oral corticosteroid-dependent asthma; as in study 1, higher exposures are probably required to achieve clinically relevant improvements in efficacy outcomes. These results concur with previous studies of the relationship between exposures and clinical efficacy outcomes for anti-IL5 therapies in patients with severe oral corticosteroid-dependent asthma.²² Results from study 2 were not pooled

with study 1 in the exposure–response analysis because of the differences in study design.

Reslizumab rapidly decreased blood eosinophil counts to a low but detectable concentration by the first visit in both studies. Decreases in blood eosinophil counts were not sufficient for clear improvements in clinical outcomes in either study. Although baseline blood eosinophil levels of 400 cells per μL or more are an adequate biomarker to indicate baseline eosinophilic airway inflammation, reduction in blood eosinophils with treatment is a surrogate biomarker, which might not fully reflect changes in eosinophilic airway inflammation. Blood eosinophil counts also decreased in the placebo groups, but to a much smaller extent. This post-baseline decrease in the placebo group might be due to regression to the mean based on the definition used for baseline eosinophil counts, which were assessed as the highest pre-dose value during screening and run-in. The decrease in peripheral blood eosinophils in placebo-treated patients might also indicate better adherence to inhaled corticosteroids during the study compared with pre-study adherence.²³

Subcutaneous reslizumab once every 4 weeks was generally well tolerated. We found few treatment-related AEs and discontinuations due to AEs. The AEs reported during the studies were consistent with the expected profiles for patients with uncontrolled severe asthma in study 1 and with severe oral corticosteroid-dependent asthma in study 2, and with previous experience with intravenous reslizumab.^{9,11} Few patients developed anti-drug antibodies, which were generally transient, low titre, non-neutralising, and not related to efficacy and safety.

These studies had several limitations. The studies were adequately powered, so we believe that the negative findings for the primary endpoints in the overall populations of both studies are meaningful; however, patient numbers for the subgroups were low, which limited the statistical power to detect subgroups that might have benefited from reslizumab. Because of multiple comparisons, a multiple testing procedure was implemented, resulting in only descriptive comparisons for secondary endpoints given that the *p* value was greater than 0.05 for the comparison of the primary endpoint in both studies. The inclusion of patients with a blood eosinophil count between 300 cells per μL and less than 400 cells per μL probably weakened our ability to show a treatment effect in either study, because a blood eosinophil count of 400 cells per μL or more appears to be a better indicator of eosinophil-driven asthma responsive to anti-IL5 therapy. A low asthma exacerbation rate in the placebo group also decreased our ability to show a treatment effect for asthma exacerbation reduction. On the basis of our post-hoc exposure–response analyses, doses of subcutaneous reslizumab higher than 110 mg once every 4 weeks will probably be required to achieve more robust improvements in efficacy outcome measures. Additionally, no long-term safety data beyond 52 weeks for subcutaneous reslizumab are available.

In conclusion, reslizumab given as a fixed-dose (110 mg) subcutaneous injection was not effective in reducing exacerbation rates in patients with uncontrolled asthma and an increased blood eosinophil count (≥ 300 cells/ μL), or in reducing the daily maintenance oral corticosteroid dose in patients with oral corticosteroid-dependent severe eosinophilic asthma. Subcutaneous reslizumab showed evidence of pharmacological activity in study 1, in terms of clinically significant improvements in lung function at 2 weeks, which were sustained until week 52, and clinically important improvements in exacerbation frequency in the subgroup of patients with high blood eosinophil counts (≥ 400 cells/ μL). However, the magnitude of effect of fixed-dose subcutaneous reslizumab on rates of clinical asthma exacerbations was lower than the 50–59% reduction observed with the higher reslizumab exposures associated with weight-based dosing of intravenous reslizumab. Reslizumab was generally well tolerated and no new safety concerns were detected. Exposure-response analyses indicate that to achieve the magnitude of effect on clinical asthma exacerbations observed with intravenous reslizumab 3 mg/kg, systemic reslizumab concentrations higher than those achieved with subcutaneous 110 mg are required. The use of higher, weight-based doses of subcutaneous reslizumab should be assessed in future studies.

Contributors

JCV, MG, and JM contributed to the conception and design of the study. JAB, JCV, KM, JFM, JJ, YA, MH, MC, JM, MG, RV, and GB contributed to the collection of data. JAB, JCV, KM, JFM, JJ, YA, MH, MC, DAM, JM, LH, MG, RV, and GB contributed to the data analysis. All authors contributed to the interpretation of the data and to the development of the first draft of the article. Subsequent drafts were reviewed and approved by all authors.

Declaration of interests

JAB has served as a speaker and consultant for AstraZeneca, BioCryst Pharmaceuticals, CSL Behring, KalVista Pharmaceuticals, Merck, Novartis, Pharming Technologies, Sanofi and Regeneron, and Shire; and has received research grants from AstraZeneca, BioCryst Pharmaceuticals, CSL Behring, Merck, Novartis, Pharming Technologies, Sanofi and Regeneron, and Shire. JCV has served as a speaker and consultant for ALK, Allergopharma, Altana, AstraZeneca, Avontec, Bayer, Bencard, Bionorica, Boehringer Ingelheim, Chiesi, Cilag, Cipla, Essex Chemie, GlaxoSmithKline, Hexal-Sandoz, Janssen, Laboratorios Leti, Meda Pharmaceuticals, Merck Sharp & Dohme, Mundipharma, Novartis, Nycomed, Pfizer, Regeneron, The Paul Ehrlich Institute, Revotar Biopharmaceuticals, Sanofi, Schering-Plough, Stallergenes, Takeda, Teva Pharmaceuticals, UCB-Schwarz Pharma, and Zydus Cadila; and has received research grants from Deutsche Forschungsgemeinschaft, Land Mecklenburg-Vorpommern, GlaxoSmithKline, and Merck Sharp & Dohme. KM has served as a speaker and consultant for AstraZeneca, Boehringer Ingelheim, Genentech, Greer, Meda, Merck, Mylan, Novartis, and Teva Pharmaceuticals. JFM has served as a speaker and consultant for AstraZeneca, ImmunoTek, Menarini, Novartis, Sanofi, Teva Pharmaceuticals, and Uriach; and has received research grants from Novartis. JJ has served as a speaker and consultant for AstraZeneca, CSL Behring, Novartis, Regeneron, Sanofi-Genzyme, Shire, and Teva Pharmaceuticals; and has received research grants from BioCryst Pharmaceuticals and Genentech. MH has served as a speaker and consultant for AstraZeneca, GlaxoSmithKline, Novartis, Roche, Sanofi, and Teva Pharmaceuticals. MC has served as a speaker and consultant for AstraZeneca, Genentech, GlaxoSmithKline, Novartis, Regeneron, Sanofi-Aventis, Teva Pharmaceuticals, Theravance Biopharma, and

VIDA; has received research grants from ALA Pharma, AstraZeneca, Chiesi, GlaxoSmithKline, the US National Institutes of Health, Novartis, the Patient-Centered Outcomes Research Institute, and Sanofi-Aventis; and receives royalties from Elsevier. GB has served as a speaker and consultant for AstraZeneca, Boehringer Ingelheim, Chiesi, GlaxoSmithKline, Novartis, Pfizer, Sanofi and Regeneron, Teva Pharmaceuticals, Union Chimique Belge, and Zambon. DAM, MG, and RV are employees of Teva Pharmaceuticals. JM and LH are former employees of Teva Pharmaceuticals. YA declares no competing interests.

Data sharing

Researchers can request access to patient-level data and related study documents including the study protocol and the statistical analysis plan. Requests will be reviewed for scientific merit, product approval status, and conflicts of interest. Patient-level data will be de-identified and study documents will be redacted to protect the privacy of trial participants and to protect commercially confidential information. Please email usmedinfo@tevapharma.com to make your request.

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